QP CODE:24900146 MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION
NOVEMBER 2024
First Semester, Discipline Specific Core Course - MG1DSCICH100 - FUNDAMENTALS OF INDUSTRIAL CHEMISTRY, (2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S) ***Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.***

Part A, Each question carries one mark.

1. Greater [U]-1
2. By using a more selective catalyst [An] [1, 2]
3. Tetraethyl lead [K] [2, 4]
4. Gamma alumina[A] [1, 2, 3]
5. To reduce the size of the ore particles [K] [3]
6. Increases corrosion [U] [1]
7. All of the above [U] [1]
8. A technique that deposits thin films from vapour phase precursors [U] [5]
9. Carbon nanotubes [K] [5]
10. Enhanced mechanical and thermal properties [A] [1, 2, 6] (10x1=10)

Part B. Short answer type, Answer any four, Each question carries three marks

11.Explain why safety measures are important in a chemical manufacturing plant [U]1,2

Any three relevant points with explanation -3 marks.

Protect Workers,PreventAccidents,Protect the Environment,SaveCosts,Follow Laws.

12. Explain how industrial chemistry bridges the gap between classical chemistry and its commercial application.U[1]

Any three differences between industrial and classical chemistry 2 marks; and applications-1 mark.

13. What are the main differences between commodity chemicals and fine chemicals?U[1]

Out of syllabus . Give credit to all

14.What is the primary structure of cellulose?K[1,2,4]

Structure -1 mark , Explanation 2 marks

15. What types of nanomaterials can be synthesized using CVD? Provide two examples.U[5]

Explanation of CVD-1 mark;Types and explanation-1D and 2D-1 mark ;Egs-CNT and Graphene -1 marks

16.How would you apply pollution control technologies to reduce emissions in a chemical manufacturing plant?A[1,2]

Waste Treatment: Processes for treating and managing waste products, such as chemical treatment, incineration, or recycling. **Emission Control Technologies**: Technologies like scrubbers, filters, and catalytic converters to capture and neutralize pollutants before release into the environment.**Sustainable Practices**: Implementing practices to reduce resource consumption, energy usage, and waste generation.**Regulatory Compliance**: Adhering to environmental regulations and standards to minimize environmental impact.

Any three methods-full marks

Part C

Short Essay Type questions , Answer any Fourquestions.Each question carries seven marks

17.Evaluate impact of isomerization on fuel quality & engine performance.A[1,2]

Definition of isomerization – 2 mark; Process explanation- 5 mark

The process uses a catalyst, typically at lower temperatures than reforming, to promote the rearrangement of hydrocarbon molecules.

o For instance, normal butane (a straight-chain alkane) can be converted into isobutane (a branched isomer), which has a much higher octane rating.

o This reaction improves gasoline's performance by making it more resistant to knocking ensuring smoother engine operation.

o The process is also environmentally beneficial, as it produces fuels that burn more cleanly.

18. "Analyze how fluctuations in raw material prices and energy costs affect the economics of a chemical process.What strategies can be used to mitigate these impacts?"An[1,2]

Beyond the scope of syllabus. Give credit for any reasonable attempt

The Economics of Chemical Processes involves analyzing and optimizing the financial aspects of chemical manufacturing to ensure cost-effectiveness and profitability which includes

Cost Estimation associated with raw materials, labor, equipment, and overheads;Economic Feasibility Studies of chemical process , return of investment; Process Optimization, Profitability Analysis:

Eg Petrochemical Industry: Conducting cost analysis to determine the most economical methods for producing and refining petrochemicals, considering factors like raw material costs and energy consumption.

Stratagies used:

1. **Use Alternatives**: Find cheaper or alternative raw materials and energy sources.
2. **Improve Efficiency**: Upgrade technology to save energy and reduce waste.
3. **Secure Prices**: Sign long-term contracts or use financial tools to lock in prices.
4. **Recycle and Reuse**: Reclaim materials to reduce dependency on new inputs.
5. **Adjust Pricing**: Increase product prices if necessary to cover costs.

Economics of chemical process -3 marks Any four Strategies used -4 marks

19.Evaluate the diverse applications of mica in modern industries and assess how its properties influence these uses.U[1,2,3]

Mica , structure – 1 mark ;types of mica -2 marks; Any two Application and Properties of mica- 4 marks

Mica is a versatile group of phyllosilicate minerals ,Forms and Structure: sheet-like structure, which is a result of the bonding between silicon-oxygen tetrahedra

Common Types of Mica: Muscovite: Light-colored, typically ranging from colorless to pale green or brown. Muscovite has high transparency and is often used in applications requiring clear insulation and electrical properties and Biotite: Dark-colored, commonly black, dark brown, or green. Biotite is more opaque and has a higher density compared to muscovite..

Applications:Electrical Insulation:, Construction,Cosmetics:Plastics and Rubber:

20.Explain the Kroll process for titanium extraction.U[1.3]

The Kroll process is a widely used method for extracting titanium metal from its ore, titanium dioxide (TiO_2), typically in the form of rutile or ilmenite.

Step 1: conversion of TiO2 to Ti Cl4. :TiO_2 is reacted with chlorine gas (Cl_2) in the presence of coke (carbon) at high temperatures ,$TiO_2+2Cl_2+2C \rightarrow TiCl_4+2CO$

Step 2:Reduction of $TiCl_4$ to Titanium Metal: $TiCl_4$ is reacted with magnesium (Mg) or sodium (Na) in an inert atmosphere. $TiCl_4+2Mg \rightarrow Ti+2MgCl_2$

Krolls Process-2 marks;Steps -2 marks;Equations -3 marks

21.Explain the importance of pollution control measures in industrial processes and how they contribute to sustainable development.U[1,2]

Definition Pollution Control- 1 mark;Importance -2 marks;Conrol measures- 2 marks

Contribution any relevant points-2 marks

1.Protecting Human Health:2. Safeguarding the Environment: 3. Complying with Regulations: 4. Reducing Waste and Improving Efficiency: 5. Supporting Sustainable Development:

22. Classify nanomaterials based on their dimensions and by citing suitable examples.U[5]

Definition of nanomaterials -2 marks.Classification with explanation – 3 marks,Examples- 2 marks

1. Zero-Dimensional (0D) Nanomaterials:In 0D nanomaterials, all three spatial dimensions are within the nanoscale. These materials are typically nanoparticles, where the material size is confined in all directions.Example: o Gold Nanoparticles (AuNPs): Spherical gold particles used in drug delivery, imaging, and catalysis.

2. One-Dimensional (1D) Nanomaterials:In 1D nanomaterials, one dimension is outside the nanoscale, but the other two dimensions remain confined to the nanoscale. These materials include nanowires, nanorods, and nanotubes.Example:o Carbon Nanotubes (CNTs):

3. Two-Dimensional (2D) Nanomaterials:In 2D nanomaterials, two dimensions are outside the nanoscale, but one dimension (thickness) is confined to the nanoscale. These materials typically exist as thin layers or sheets.Example: o Graphene:

4. Three-Dimensional (3D) Nanomaterials:In 3D nanomaterials, none of the dimensions are confined to the nanoscale, but they can contain internal structures or assemblies that exhibit nanoscale features. These include bulk materials with nanoscale porosity or composites with nanostructured components.Example: Nanoporous Materials: Materials with nanoscale pores, used in catalysis, gas storage, and separation technologies.